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Non-iterative phase tilt interferometry for the tilt shift calculation and phase extraction

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Abstract: This study proposed a non-iterative phase tilt interferometry (NIPTI) for the tilt shift calculation and phase extraction, which can solve the random tilt-shift problem caused by external vibration. The method approximates the higher order terms of the phase to make it suitable for linear fitting. Then according to an estimated tilt, the accurate tilt shift can be obtained by least squares method without iteration, and the phase distribution can be calculated by it. The simulation results indicated that the root mean square error of the phase calculated by the NIPTI can reach up to 0.0002λ. The experimental results showed that the calculated phase exhibited no significant ripple when using the NIPTI for cavity measurements in a time-domain phase shift Fizeau interferometer. Moreover, the root mean square repeatability of the calculated phase reached up to 0.0006λ. It indicates that the NIPTI provides an efficient and high-precision solution for random tilt-shift interferometry under vibration.

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1. Introduction

In phase shift interferometry, the phase shift is used as a known value for the phase extraction. In practice, the phase shift deviates from the theoretical value owing to certain factors such as external vibration or nonlinear error of the phase shifter, leads to phase error. To solve this problem, scholars have proposed the random phase-shifting algorithm, which was developed together with the study of the error sensitivity of the traditional fixed-step phase-shifting algorithm. Typically, the random phase-shifting algorithm employs an iterative method to extract the phase. The most representative one is the advanced iterative algorithm (AIA) proposed by Wang [1], which only requires more than three interferograms to extract the phase. AIA has no requirement in terms of the phase shift and exhibits high accuracy. However, it also has some problems. On the one hand, a multitude of least squares operations in the iterative process decrease the calculation efficiency, and on the other hand, it is prone to error under the few-fringes condition. Until 2011, Vargas [2–4] applied the principal component analysis (PCA) in mathematical statistics to interferogram analysis. PCA has low computational cost compared to the AIA. Unfortunately, the PCA often gives erroneous phase demodulation so subsequent works proposed to patch it by adding few AIA iterations to obtain a much better phase estimation [5–8]. The above shows that the serious disadvantage of both AIA and PCA is that quite often, the demodulated phase may have unacceptable errors and there is no way to be aware of this circumstance. In this regard, Servin [9] proposed a universal phase-shifting algorithm (UPSA), which efficiently solves the limitations presented by previous phase-shifting algorithms. It works for any linear, nonuniform, or even random unknown phase steps. Additionally, Servin proposed the modulus of the demodulated signal as null-test criterion for easily spotting an erroneous demodulated phase from experimental interferograms.

The above studies only focused on random translational phase shift, that is, the phase shift was only a function of time and its spatial distribution was consistent. Whereas in practice, the

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tilt variation of the reference and test mirrors is inevitable, such as the tilt wobble of the lens caused by the external vibration, the tilt of the reference mirror caused by the inconsistency of the multi-point PZT step. In the process of solving the tilt-shift problem, on the one hand, some of the ideas in the above methods have been continued. For example, Xu [10] improved on the AIA to calculate the phase shift by creating chunks of the interferograms and constructing a plane fit to realize the calculation of the tilt shift. This method exhibited less error when the background and contrast of interferograms were uniform; however, when the two were unevenly distributed, it resulted in coupling effect on the fitting of the tilt coefficient [11]. Meanwhile, some of the issues in the AIA were passed down, such as the low efficiency, unacceptable errors under the few-fringes condition. On the other hand, scholars have proposed some new methods, and it also raises some new problems. Chen [12] first proposed a method to solve the tilt-shift problem. It based on a first-order Taylor expansion of interferograms and used least-square iterations to achieve small-amplitude tilt shift error compensation. In 2013, Li [13] solved the tilt-shift based on the concept of detecting straight lines via Hough transform [14] to realize the phase extraction in random tilt-shift interferograms. It has a same problem to the classical methods, which is prone to error under the few-fringes condition. In 2014, Deck [15] proposed a method called model-based phase shifting interferometry (MPSI). It constructed a multi-parameter physical model, which can't be solved directly. Therefore, the method is to linearize the model and apply iterative least-squares techniques. It used the background and contrast as time-independent quantities to compensate for the tilt shift in the phase. In the same year, Juarez-Salazar [16] proposed a random phase-shifting algorithm in which the phase shift, background intensity, and amplitude were variable in both the time and space dimensions. It greatly improved the universality of the random phase-shifting algorithm. However, the method requires the interferograms has a large number of fringes. In 2021, Lu [17] calculated the tilted phase plane of each interferogram based on a Fourier transform to realize the phase extraction. Similarly, it needs a large number of fringes in the interferograms to avoid the spectral overlap. Then, Duan [18] proposed an iterative method for solving tilted phase-shifted interferograms. This method constructed a linear system of equations about the tilt parameters of the interferogram to solve the phase shift between interferograms, and then realized the phase extraction.

In summary, there are still some problems with the random phase shifting algorithms, which are used to solve the tilt shift problem, such as low computational efficiency caused by the iterative, unacceptable errors under the few-fringes condition or the uneven fringes' background and contrast condition or the large-amplitude tilt shift condition. Therefore, this study proposed a non-iterative phase tilt interferometry for the tilt shift calculation and phase extraction. It is highly efficient due to the non-iteration, and there is no excessive requirement regarding the number of fringes. The tilt shift between interferograms can be randomly varied at will without any effect. Last but not least, the accurate phase distribution can be obtained under the condition that the background and contrast of interferograms are uneven.

2. Theoretical principles

The intensity of a single-frame interferogram can be expressed as

$$I = a + b\cos(\varphi_0 + mx + ny + k) \tag{1}$$

where a and b are the background and amplitude of the interferogram, respectively, m and n are the first-order term coefficients of the phase, corresponding to the x and y components, respectively, k is the constant term coefficient of the phase, and φ_0 is the higher-order component of the phase above the first-order.

For a single-frame interferogram, m, n, and k are collectively referred to as the tilt parameters of the interferogram. Based on the Fourier transform, the tilt parameters m and n can be easily

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estimated and referred to as the initial values of the tilt parameters, denoted as m_0 and n_0. Thus, Eq. (1) can be rewritten as follows

$$I = A_k + B_k\cos(mx+ny) + C_k\sin(mx+ny), with \begin{cases} A_k = a \\ B_k = b\cos(\varphi_0 + k) \\ C_k = -b\sin(\varphi_0 + k) \end{cases} \tag{2}$$

Considering that in practice $\varphi_0 << m, n, k$, φ_0 was omitted. According to the initial values of the tilt parameters m_0 and n_0, the least-square form can be constructed with parameters A_k, B_k, and C_k. The variance between the theoretical intensity and the actual intensity can be expressed as

$$E = \sum (A_k + B_k\cos(mx+ny) + C_k\sin(mx+ny) - I)^2 \tag{3}$$

To minimize the variance, the following solution is obtained.

$$P \cdot \begin{bmatrix} A_k & B_k & C_k \end{bmatrix}^T = \begin{bmatrix} \sum_{i=1}^{M}\sum_{j=1}^{N} I_{ij} & \sum_{i=1}^{M}\sum_{j=1}^{N} I_{ij}\cos(\alpha) & \sum_{i=1}^{M}\sum_{j=1}^{N} I_{ij}\sin(\alpha) \end{bmatrix}^T, with$$
$$P = \begin{bmatrix} M\cdot N & \sum_{i=1}^{M}\sum_{j=1}^{N}\cos(\alpha) & \sum_{i=1}^{M}\sum_{j=1}^{N}\sin(\alpha) \\ \sum_{i=1}^{M}\sum_{j=1}^{N}\cos(\alpha) & \sum_{i=1}^{M}\sum_{j=1}^{N}\cos^2(\alpha) & \sum_{i=1}^{M}\sum_{j=1}^{N}\sin(\alpha)\cos(\alpha) \\ \sum_{i=1}^{M}\sum_{j=1}^{N}\sin(\alpha) & \sum_{i=1}^{M}\sum_{j=1}^{N}\sin(\alpha)\cos(\alpha) & \sum_{i=1}^{M}\sum_{j=1}^{N}\sin^2(\alpha) \end{bmatrix} \tag{4}$$

where $\alpha = mi + nj$, M and N are the number of rows and columns of the interferogram, respectively, and I_{ij} is the intensity value of row i and column j in the interferogram.

After obtaining the parameters A_k, B_k, and C_k, the constant term coefficient k can be expressed as

$$k = \tan^{-1}(-C_k/B_k) \tag{5}$$

However, there is inevitably an error between the initial value m_0 and n_0 and their actual value. Regardless, the constant term coefficient k obtained by this method is still very accurate.

After obtaining the exact constant term coefficient k, Eq. (1) can be rewritten as

$$I = A_m + B_m\cos(ny) + C_m\sin(ny), with \begin{cases} A_m = a \\ B_m = b\cos(\varphi_0 + mx + k) \\ C_m = -b\sin(\varphi_0 + mx + k) \end{cases} \tag{6}$$

Similar to the previous case, φ_0 is omitted.

$$\begin{bmatrix} M & \sum_{i=1}^{M}\cos(ni) & \sum_{i=1}^{M}\sin(ni) \\ \sum_{i=1}^{M}\cos(ni) & \sum_{i=1}^{M}\cos^2(ni) & \sum_{i=1}^{M}\sin(ni)\cos(ni) \\ \sum_{i=1}^{M}\sin(ni) & \sum_{i=1}^{M}\sin(ni)\cos(ni) & \sum_{i=1}^{M}\sin^2(ni) \end{bmatrix} \begin{bmatrix} A_m \\ B_m \\ C_m \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{M} I_{ij} \\ \sum_{i=1}^{M} I_{ij}\cos(ni) \\ \sum_{i=1}^{M} I_{ij}\sin(ni) \end{bmatrix} \tag{7}$$

According to the initial value n, the data of a column in the interferogram (such that mx is a constant value) was considered to construct the least-square form with parameters A_m, B_m, and C_m, as shown in Eq. (7).

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After obtaining the parameters A_m, B_m, and C_m, the first order term coefficient m can be expressed as

$$mx + k = \tan^{-1}(-C_m/B_m) \tag{8}$$

For the same reason, the-least squares form for solving the first-order term coefficient n can be constructed by considering a row of data in the interferogram (such that ny is a constant value) according to the initial value m_0 (or the exact m calculated from the initial value n_0). The form of the solution is similar to that of Eq. (7), provided n in Eq. (7) is replaced by m, as shown in Eq. (9).

$$\begin{bmatrix} N & \sum_{j=1}^{N}\cos(mj) & \sum_{j=1}^{N}\sin(mj) \\ \sum_{j=1}^{N}\cos(mj) & \sum_{j=1}^{N}\cos^2(mj) & \sum_{j=1}^{N}\sin(mj)\cos(mj) \\ \sum_{j=1}^{N}\sin(mj) & \sum_{j=1}^{N}\sin(mj)\cos(mj) & \sum_{j=1}^{N}\sin^2(mj) \end{bmatrix} \begin{bmatrix} A_n \\ B_n \\ C_n \end{bmatrix} = \begin{bmatrix} \sum_{j=1}^{N} I_{ij} \\ \sum_{j=1}^{N} I_{ij}\cos(mj) \\ \sum_{j=1}^{N} I_{ij}\sin(mj) \end{bmatrix} \tag{9}$$

After obtaining the parameters A_n, B_n, and C_n, the first order term coefficient m can be expressed as

$$ny + k = \tan^{-1}(-C_n/B_n) \tag{10}$$

In summary, for a single-frame interferogram, firstly, the parameter k can be fitted by the estimated parameters m_0, n_0. Secondly, the parameter m can be fitted by the estimated parameter n_0 and the parameter k already obtained in the previous step. Finally, parameter n can be fitted by the parameter m obtained in the previous step and the parameter k obtained in the first step. It shows that the new estimations are passed down as initial guesses for the demodulation of the following fringe pattern. However, in the NIPTI, each parameter is fitted only once, and the accurate parameters can be obtained from this single fitting. Thus, for the tilt shift interferograms, as long as the tilt parameters of the interferograms are obtained frame by frame, the tilt shift calculation or phase extraction can be achieved. The flowchart of the NIPTI is shown in Fig. 1.

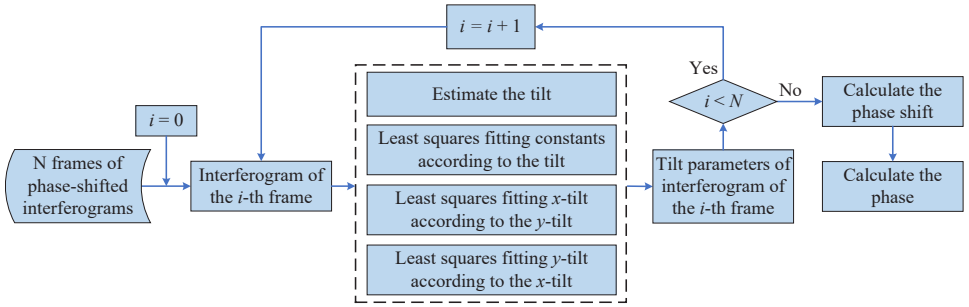


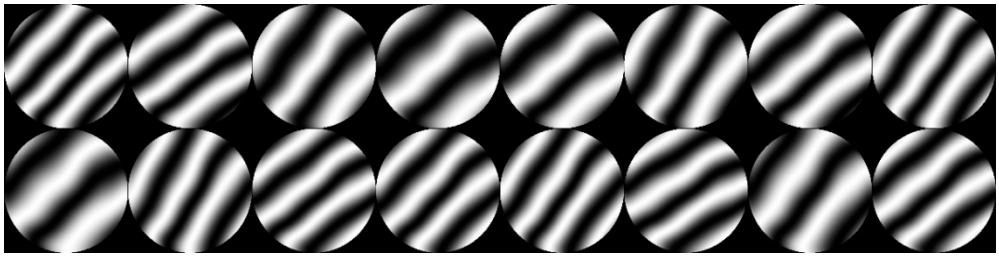
Fig. 1. The flowchart of the NIPTI.

3. Simulation

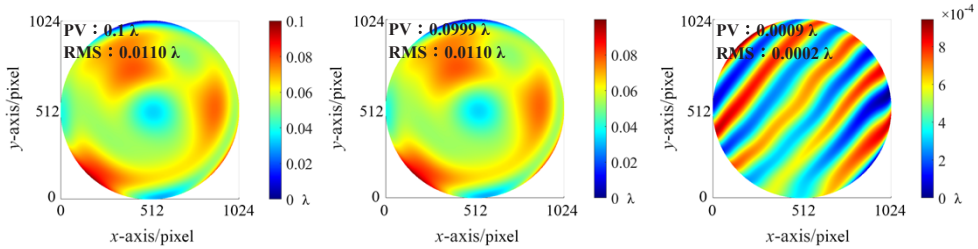
To test the performance of the NIPTI, the phase is required, which was generated by Zernike polynomial and its PV was 0.1λ, as shown in Fig. 2(b). The tilt and phase shifts were added to generate phase-shifted interferograms, where the phase shift was divided into two parts: equally spaced phase shift of π/8 rad and random noise. The noise contained both the constant error generated by the translation and the first order term error generated by the tilt. The amplitude of the constant with time was π/16 rad, the amplitude of the tilt with time was 1λ, and the number

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of fringes in interferograms was 2–4. A total of 16 frames of phase-shifted interferograms were acquired, as shown in Fig. 2(a).



(a) Phase-shifted interferogram



(b) Actual phase (c) Calculated phase (d) Residual

Fig. 2. Simulation results of phase calculation by the NIPTI.

The tilt parameters m, n, and k of each interferogram were calculated using the NIPTI. Further, the difference was incorporated to obtain the tilt parameter of the phase shift, which were denoted as δ_m, δ_n, and δ_k. The error between these values and the actual tilt parameters of the phase shift were denoted as $\Delta\delta_m$, $\Delta\delta_n$, and $\Delta\delta_k$, as presented in Table 1.

Table 1. The tilt parameter of the phase shift calculated by the NIPTI and its error

Interferograms number	δ_m / λ	δ_n / λ	δ_k / λ	$\Delta\delta_m$ / λ	$\Delta\delta_n$ / λ	$\Delta\delta_k$ / λ
1	/	/	/	/	/	/
2	0.3775	0.0417	0.0243	-0.0021	0.0010	-0.00024
3	0.0770	0.5355	0.0241	0.0034	-0.0030	0.00055
4	0.2985	0.0601	0.0311	0.0047	0.0016	-0.00053
5	0.0980	0.1317	0.0194	0.0058	-0.0044	0.00045
6	0.3537	0.2274	0.0276	-0.0105	0.0002	-0.00035
7	0.2430	0.3468	0.0184	-0.0060	-0.0021	-0.00023
8	0.4274	0.1368	0.0200	0.0016	-0.0027	0.00042
9	0.5825	0.1196	0.0400	-0.0027	-0.0006	-0.00006
10	0.6468	0.0238	0.0250	-0.0059	-0.0011	0.00012
11	0.3351	0.5193	0.0339	-0.0012	-0.0028	-0.00029
12	0.1246	0.1712	0.0458	-0.0006	-0.0011	0.000001
13	0.2135	0.1541	0.0287	-0.0059	-0.0011	-0.00020
14	0.5277	0.3299	0.0379	-0.0060	-0.0028	0.00028
15	0.0955	0.6025	0.0387	0.0005	-0.0036	-0.00031
16	0.0383	0.4799	0.0296	0.0006	-0.0012	0.00022

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Based on the calculated phase shift, the phase can be obtained, as shown in Fig. 2(c), and the residual of the phase calculated by the NIPTI and its nominal value is shown in Fig. 2(d).

For the same phase, the tilt shift was randomly generated to obtain the phase-shifted interferograms. Subsequently, the phase was solved using the NIPTI and its root mean square was calculated. This was repeated for 36 groups to obtain the root mean square repeatability of the NIPTI, as presented in Table 2.

Table 2. Simulation of root mean square repeatability of the NIPTI

Number	RMS/λ	Number	RMS/λ	Number	RMS/λ
1	0.01101	13	0.01101	25	0.01101
2	0.01100	14	0.01099	26	0.01099
3	0.01099	15	0.01096	27	0.01101
4	0.01101	16	0.01100	28	0.01099
5	0.01099	17	0.01100	29	0.01101
6	0.01099	18	0.01096	30	0.01101
7	0.01100	19	0.01101	31	0.01099
8	0.01101	20	0.01098	32	0.01100
9	0.01100	21	0.01101	33	0.01101
10	0.01100	22	0.01101	34	0.01100
11	0.01103	23	0.01099	35	0.01103
12	0.01101	24	0.01101	36	0.01101
Root mean square repeatability				0.000015λ	

The above simulation shows that when the NIPTI was used to calculate the tilt parameter of the interferogram, the error of the first order term reached up to the order of $10^{-3}\lambda$, whereas that of the constant reached up to the order of $10^{-4}\lambda$. The root mean square error (RMSE) of the phase calculated by it reached be up to 0.0002λ, and the root mean square repeatability reached up to 0.000015λ. Further, the theoretical performance of the NIPTI was good and thus it could be used for the calibration of tilted phase-shifted interferograms and phase calculation under the vibration condition.

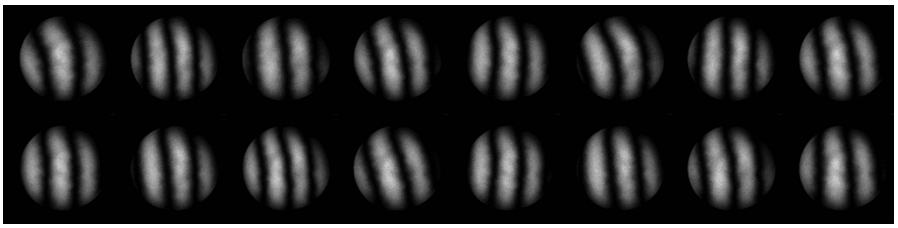
4. Experiment

To verify the feasibility of the NIPTI, experiments were conducted on a 32-inch horizontal Fizeau interferometer. The PV of the relative surface distribution between the reference surface and the test surface can reach λ/10. Meanwhile, the reference and test mirrors tilt change with time owing to external vibration, rendering the number and direction of fringes in the interferograms to vary randomly, which complicates the phase calculation. The random tilt shift interferograms collected by the interferometer are shown in Fig. 3(a). As a comparison, the standard phase-shifted interferograms were collected under vibration isolation conditions, as shown in Fig. 3(b).

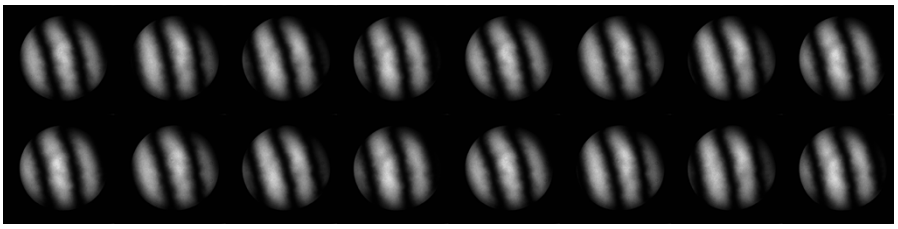
For the random tilt shift interferograms shown in Fig. 3(a), four methods are used to extract the phase, which are the four-step method, AIA, PTI, and NIPTI, as shown in Figs. 4(a), (b), (c), (d). The standard phase obtained from the standard phase-shifted interferograms shown in Fig. 3(b) is shown in Fig. 4(e). The residuals between the phase solved by the PTI or NIPTI and the standard phase are shown in Figs. 4(f), (g). Finally, the background and contrast of the interferograms shown in Fig. 3(a) are solved by the NIPTI, as shown in Figs. 4(h), (i).

Figures 4(a), (b) verify that the four-step method and AIA are not suitable for the demodulation of the tilt shift interferograms, with the solved phase mainly yielding ripple errors. Figures 4(d), (c)

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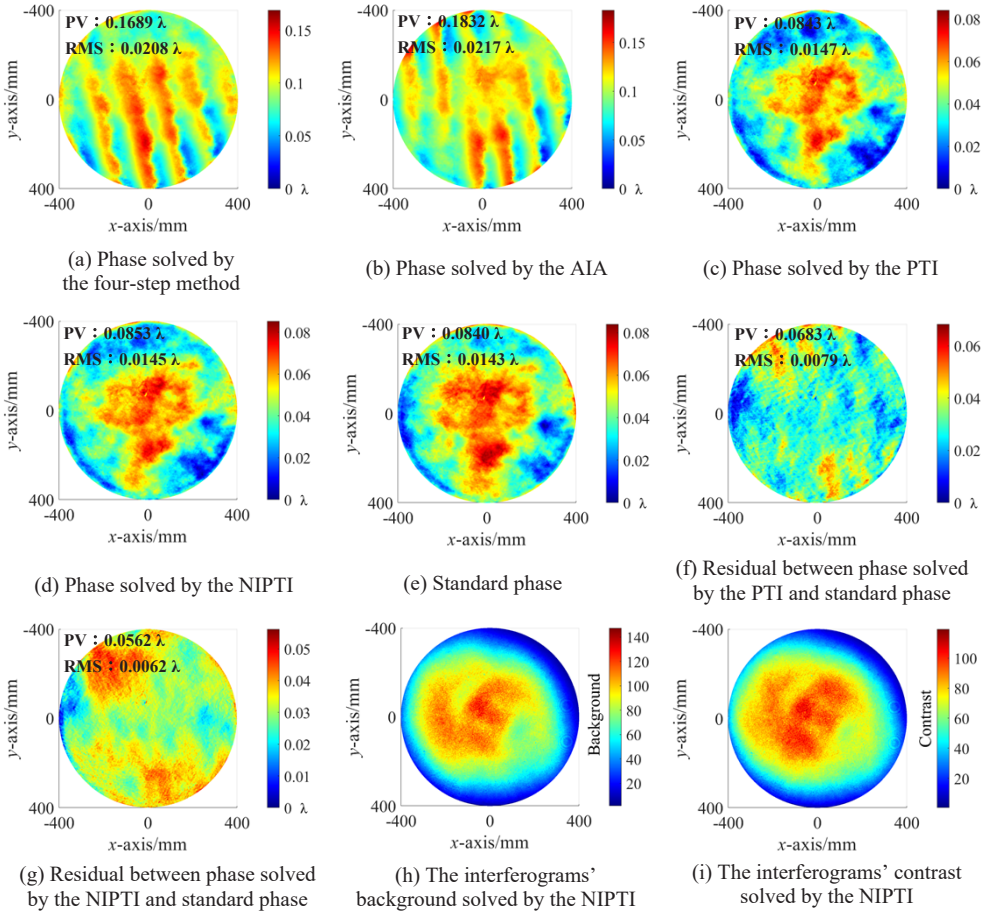


(a) The random tilt shift interferograms



(b) The standard phase shift interferograms

Fig. 3. Interferograms collected under different conditions.



(a) Phase solved by the four-step method

(b) Phase solved by the AIA

(c) Phase solved by the PTI

(d) Phase solved by the NIPTI

(e) Standard phase

(f) Residual between phase solved by the PTI and standard phase

(g) Residual between phase solved by the NIPTI and standard phase

(h) The interferograms' background solved by the NIPTI

(i) The interferograms' contrast solved by the NIPTI

Fig. 4. Experimental results on the comparison of the NIPTI for phase extraction.

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show that both PTI and NIPTI can be used for the phase extraction of tilt shift interferograms. However, the comparison between Fig. 4(f) and Fig. 4(g) shows that the phase solved by the PTI still have a small ripple error remaining, while the phase solved by the NIPTI is closer to the standard phase and has no ripple error. Last but not least, the background and the contrast of the interferograms also can gauge the quality of the demodulated phase, as shown in Figs. 4(h), (i). It proves the accuracy of the phase solved by the NIPTI from another perspective.

To further verify the root mean square repeatability of the NIPTI for the phase calculation, the algorithm was used to solve the phase and the root mean square of the phase were calculated. This was repeated for 36 groups to calculate the root mean square repeatability of the phase calculated by the NIPTI, as presented in Table 3.

Table 3. Experiment results of root mean square repeatability of phase calculation using the NIPTI

Number	RMS/λ	Number	RMS/λ	Number	RMS/λ
1	0.0130	13	0.0139	25	0.0134
2	0.0139	14	0.0128	26	0.0135
3	0.0156	15	0.0129	27	0.0142
4	0.0140	16	0.0130	28	0.0140
5	0.0136	17	0.0131	29	0.0141
6	0.0135	18	0.0131	30	0.0133
7	0.0140	19	0.0131	31	0.0137
8	0.0139	20	0.0128	32	0.0139
9	0.0129	21	0.0137	33	0.0140
10	0.0135	22	0.0140	34	0.0134
11	0.0142	23	0.0142	35	0.0129
12	0.0140	24	0.0141	36	0.0136
Root mean square repeatability				0.00056λ	

The cavity RMS repeatability reached up to 0.00056λ when tested under vibration environment using the NIPTI. The experimental results showed that the NIPTI exhibited good vibration resistance and high stability. Further, it could be used for time-domain phase shift interferometry in complex environments.

5. Discussion

5.1. Effect of initial values on the NIPTI

The accuracy of the NIPTI is dependent on the estimated initial values of the tilt parameters m_0 and n_0. In general, there is a certain error between the initial and actual values, which can affect the method. Firstly, in order to avoid the effect of other factors, two ideal interferograms are generated, with the phase being the ideal tilting surface, as shown in Fig. 5. The tilt of the phase of each interferogram is known, and an error is added as the initial value, with a variation range of -0.5λ to 0.5λ. This initial value is brought into NIPTI to calculate the tilt parameter m, n, and k of the interferograms. In practice, whether it is the calibration of the phase shifter or the calculation of the phase, the focus was on the accuracy of the phase shift. The phase shift could be described by the tilt coefficient δ_m, δ_n, and δ_k, so the tilt coefficient of the phase shift can be obtained by making a difference between the tilt parameters of the two interferograms. The error of this tilt coefficient from its nominal value is noted as $\Delta\delta_m$, $\Delta\delta_n$, and $\Delta\delta_k$. Their variation curves with the errors of the initial values are shown in Figs. 6(a), (b), (c).

As evident in Fig. 6, when the error of the initial value was within 0.5λ, the error of the phase shift calculated using the NIPTI was on the order of 10^{-3}λ, which is within the acceptable range.

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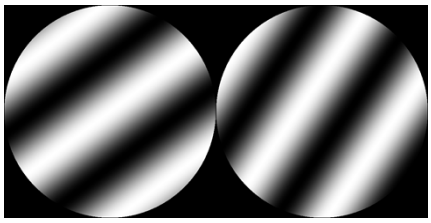


Fig. 5. The two ideal interferograms.

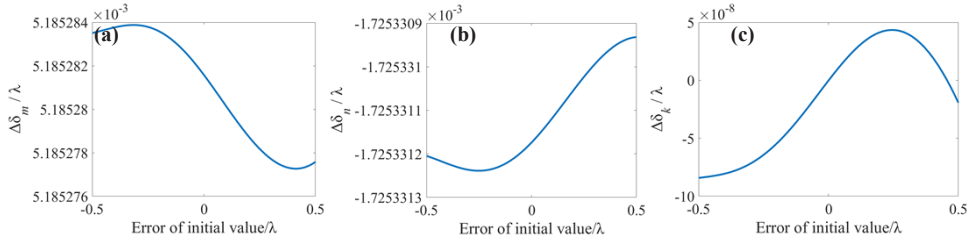


Fig. 6. Variation curve of the error of the phase shift with the error of initial value.

In the absence of any error in the initial value, the error of the tilt parameters or phase shift calculated by the NIPTI converged to a constant (instead of 0). When there was an error in the initial value, the error of the phase shift fluctuated around this constant, and the fluctuation was in the range of better than $10^{-7}\lambda$. Therefore, the NIPTI was insensitive to the error of the initial value. Meanwhile, the common method of estimating the tilt parameters of interferograms could easily control the initial value error within 0.5λ, such as the Fourier transform method, etc. Thus, the tilt parameters calculated by the NIPTI are very accurate in almost all cases. Moreover, the method is not limited by the selection of initial values, which avoids complex iterations and considerably improves the efficiency of the algorithm.

5.2. *Effect of the number of interferograms on the NIPTI*

NIPTI requires a minimum number of 3 interferograms in phase extraction, and the number of interferograms has an impact on the method. Using the phase shown in Fig. 2(b), a random phase shift is added to generate 32 frames of tilt-shift interferograms, as shown in Fig. 7. N frames of these interferograms are selected for phase extraction by the NIPTI, and the RMSE of the phase is calculated. When N gradually increases from 3 to 32, the variation curve of the RMSE of phase with the number of interferograms can be obtained, as shown in Fig. 8.

As shown in Fig. 8, the RMSE decreases as the number of interferograms approximates the inverse proportional function. In general, a RMSE of 0.01λ can be achieved as long as the number of interferograms is larger than 3. When the number of interferograms is larger than 5, the RMSE is better than 0.002λ, and when the number of interferograms is larger than 15, the RMSE tends to be stable and better than 0.001λ.

In summary, when using the NIPTI, the phase-shifted interferograms should be collected more than 5 to ensure the accuracy, and 16 frames of phase-shifted interferograms are sufficient if you want to obtain higher accuracy.

5.3. *Effect of phase φ_0 on the NIPTI*

A prerequisite exists for the use of the NIPTI, that is, the higher order term of the phase φ_0 should be far smaller than the tilt parameters m, n, and k of the phase. This condition is obviously not

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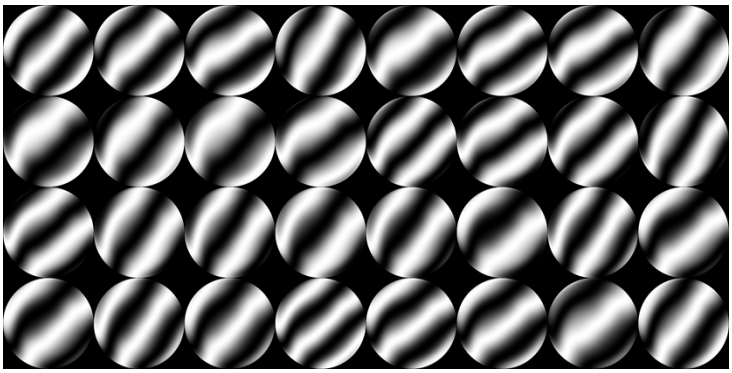


Fig. 7. 32 frames of tilt-shift interferograms.

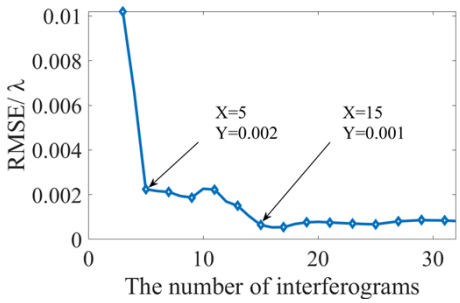


Fig. 8. Variation curve of RMSE with the number of interferograms.

satisfied when converging to the zero position measurement; thus, the phase φ_0 is contradictory to the tilt parameters m, n, and k. Therefore, the phase φ_0 is contradictory to the number of fringes of the interferogram, and when the phase φ_0 is determined, there exists a minimum number of interferogram fringes to satisfy the accuracy of the algorithm. Similarly, when the number of interferogram fringes is determined, there exists a maximum phase φ_0, where the magnitude of the phase can be evaluated using the parameter PV.

First, consider the effect of phase φ_0 on the accuracy of the NIPTI when the number of fringes is determined, which can be divided into two parts. One is the effect of the shape of φ_0, and the other is the effect of the PV of φ_0.

When the shape of φ_0 is inconsistent, it may exhibit different effects on the accuracy of the NIPTI. Therefore, φ_0 Was determined as a single aberration component, and the PV was also determined as 0.1λ. Further, the phase shifted amount containing the tilt term was added randomly to generate the phase shifted interferogram. Phase-shifted interferograms were generated via the random addition of a phase shift containing a tilt. Subsequently, the tilt parameters of the interferogram were calculated using the NIPTI, and the phase was calculated by it. This was repeated for 36 groups to calculate the RMSE curve when φ_0 was determined as a single aberration component, as shown in Fig. 9. Figures 9(a), (b), (c), (d), (e), and (f) correspond to the root mean square error curves when φ_0 was Power, Astig x, Astig y, Coma x, Coma y, and Primary Spherical, respectively.

Figure 9 shows that when the PV of φ_0 was determined, the root mean square error of the phase reached up to the order of $10^{-4}\lambda$. This indicated that the effect of φ_0 on the accuracy of the NIPTI was the same and could be ignored. Thus, the NIPTI can be applied to the phase solution of almost any surface shape, while only its PV needs to be considered.

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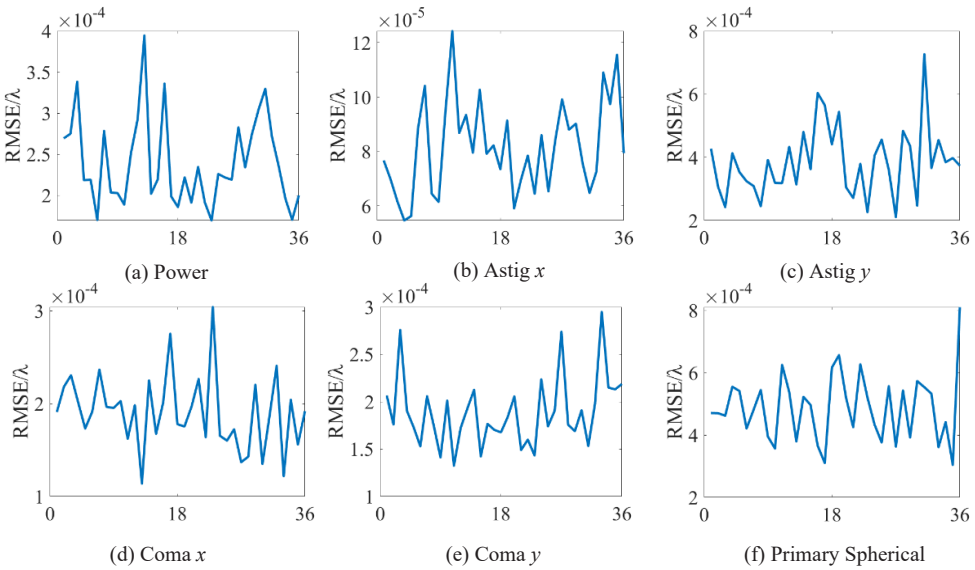


Fig. 9. RMSE of the phase calculated by the NIPTI when φ_0 determined as different aberrations

Considering PV of φ_0, the surface of φ_0 was generated by 36-term Zernike polynomial, and the tilt parameters of 16-frame interferograms were randomly generated to make the number of fringes in interferograms as 2–3. To control the variables, the PV of φ_0 was uniformly varied from 0 to 1λ using linear scaling method, and the tilted phase-shifted interferograms were constructed based on it. Thereafter, the phase shift was calculated using the NIPTI and consequently the phase was calculated using this phase shift. Thus, the variation curve of the RMSE of the phase with PV of φ_0 was obtained, as shown in Fig. 10.

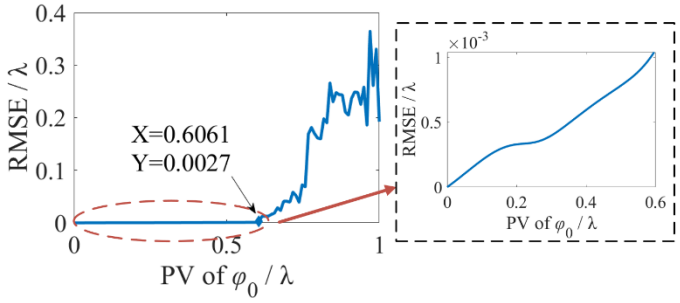


Fig. 10. Variation curve of RMSE of the phase calculated by the NIPTI with PV of φ_0

Figure 10 shows that, to satisfy the prerequisite of $\varphi_0 << m, n, k$ when calculating phase shift using the NIPTI, the PV of phase should be less than 0.6λ when the number of fringes is 2–3. In this case, the root mean square error of the NIPTI can be better than 0.001λ, and the error increases approximately linearly with the increase in PV of φ_0.

Thus, PV of φ_0 affects the accuracy of the NIPTI. When the number of fringes is determined, there exists a threshold for the PV of φ_0, beyond which the algorithm cannot be available, and within which the error is small such that it can be neglected. Meanwhile, the error increases approximately linearly with the PV of φ_0.

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5.4. Effect of the number of fringes on the NIPTI

When the PV of the measured phase was determined, the accuracy of the NIPTI was dependent on the magnitude of the phase tilt parameters m and n, that is, the number of fringes in the interferogram. The Zernike polynomial was used to generate φ_0 with PV of 0.1λ, and tilt was added to generate a single-frame interferogram. The tilt parameters m, n, and k of this interferogram were calculated using the NIPTI, whose errors with the actual values were noted as Δm, Δn, Δk. When the number of fringes varied as 0–10, the variation curves of errors Δm, Δn, Δk with the tilt of the phase were obtained, as shown in Figs. 11(a), (b), (c), respectively. The interferogram was phase-shifted at equal intervals of π/8 rad with no tilt change during the phase shift to ensure that the tilt parameters m and n were consistent for each interferogram. The phase shift was calculated using the NIPTI and the phase was calculated using this phase shift. Similarly, when the tilt of phase changed as 0–10λ, the variation curve of the RMSE of the phase with tilt of phase was obtained, as shown in Fig. 11(d).

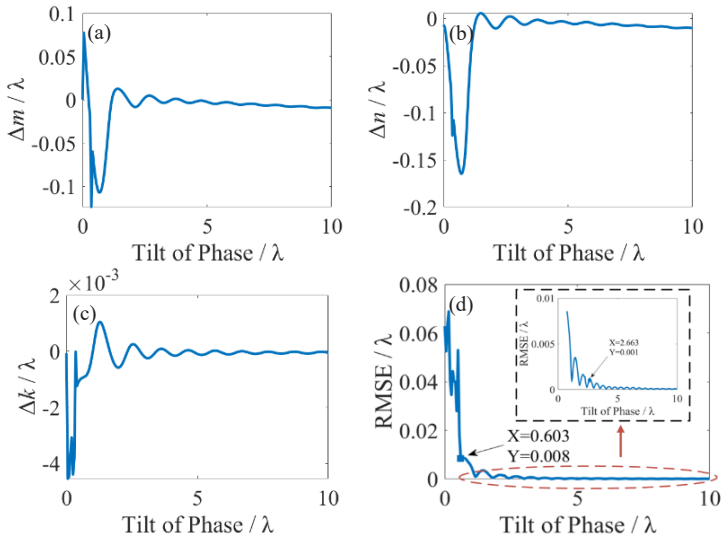


Fig. 11. Variation curve of the error of tilt parameters and phase calculated by the NIPTI with the number of fringes.

As evident in Figs. 11(a), (b), and (c), when the PV of φ_0 was determined as 0.1λ, the tilt parameters m and n calculated by the NIPTI yielded large errors when the number of fringes is small; the error of the tilt parameter k was small in almost any case, reaching up to the order of 10^{-3}λ. It indicated that if there was no tilt change in phase shift, the NIPTI could be simplified by using it only for the calculation of the constant of phase, which greatly improves the applicability of the method. Moreover, it can even be used for zero position measurements.

As shown in Fig. 11(d), the RMSE varied approximately with the number of fringes as an inverse ratio function when the phase was calculated using the NIPTI. Under the condition that the PV of φ_0 was determined as 0.1λ, the root mean square error of the phase was better than 0.008λ when the tilt of phase was more than 0.6λ. Further, the RMSE of the phase was better than 0.001λ when the tilt of phase was greater than 2.7λ.

Thus, the number of interferogram fringes affected the accuracy of the NIPTI. When PV of φ_0 is determined, there exists a threshold for the number of fringes. When the number of fringes is greater than the threshold, the NIPTI can be used. Meanwhile, the error decreases as an inverse ratio function with the number of fringes. Therefore, in practice, the interferometer should be

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adjusted with the interference phase, such that the interferogram has a reasonable number of fringes to satisfy the measurement accuracy requirements.

6. Conclusion

This study proposed a non-iterative phase tilt interferometry for the tilt shift calculation and phase extraction, which can solve the random tilt-shift problem caused by external vibration. The method is insensitive to the initial values so that it doesn't need to iterate. It indicates that NIPTI is highly efficient. The NITPI needs at least 5 frames of phase-shifted interferograms to extract the phase with guaranteed accuracy, and 16 frames of phase-shifted interferograms are sufficient to achieve the best performance. Last but not least, the NIPTI can be applied for the phase-shifted interferograms which can be randomly tilt shifted at will. The only prerequisite for applying the NIPTI is that the phase should be much smaller than the tilt. When the tilt are larger, that is, the fringes is more, and the phase is smaller, the accuracy of the method is higher.

The NIPTI can also be split and used to solve the constant or tilt of the phase separately. If the vibration during phase shifting is small and the change of tilt is negligible, the NIPTI can be simplified and used only to solve the constant of phase, which can be used to calculate the phase shift of traditional random translational phase-shifted interferograms. The NIPTI requires at least two interferograms to calculate the phase shift. Further, it has almost no requirement for the number of fringes, and high accuracy can be achieved at zero position measurement. In addition, certain scenarios require the tilt of the phase as the input parameter, such as the correction of interferometer retrace error [19], elimination of tilt error in the shift-rotation absolute measurement [20], etc. The accuracy of the input parameters can be greatly improved if the NIPTI is used. Taken together, the NIPTI has a wide range of application and can provide a highly efficient and accurate solution for phase-shifting interferometry in vibration conditions.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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